

Vitor Negromonte Cabral de Oliveira

Recife, BR contato.vnco@gmail.com linkedin.com/in/vitornegromonte github.com/vitornegromonte

Education

Federal University of Pernambuco

B.Sc Information Systems

Recife

Exp. 2028

Relevant activities: Artificial Intelligence League Co-founder, Teaching Assistant, Undergraduate Researcher.

Work Experience

Software Engineering Intern (Machine Learning)

Sinapse — Polícia Civil de Pernambuco

Dec 2025 – Present

Recife, BR

- Designed and implemented the backend architecture for an AI-powered criminal intelligence platform supporting criminal investigations.
- Built asynchronous ETL pipelines following the Medallion architecture using PostgreSQL and MinIO.
- Developed REST APIs for OCR, NER, LLM, graph neural network, and clustering services to transform heterogeneous investigative data into searchable intelligence through automated document processing and relationship analysis.
- Implemented immutable auditing and automated PII de-identification to ensure LGPD compliance.

Technologies: Python, FastAPI, PostgreSQL, vLLM, pgvector, Redis, Docker, Celery, MinIO.

Machine Learning Engineer

Freelancer

Jan 2025 – Sep 2025

Remote

- Designed and deployed an end-to-end computer vision system for agricultural applications.
- Built data collection, annotation, preprocessing, and training workflows for custom image datasets.
- Optimized deep learning models for real-time edge inference in production environments.
- Developed containerized deployment pipelines to support reproducible model delivery.

Technologies: Python, PyTorch, OpenCV, Docker, Linux, Git.

Data Scientist

redduo.ai

Nov 2023 – Jul 2024

Recife, Brazil

- Developed NLP pipelines for business intelligence applications using modern machine learning techniques.
- Built automation tools that streamlined data analysis and reporting workflows.
- Improved model performance and deployment scalability through pipeline optimization.
- Integrated cloud services to support production-ready AI applications.

Technologies: Python, PyTorch, Scikit-learn, AWS Lambda, AWS Cognito, Power BI.

Research Experience

Center for Informatics, Federal University of Pernambuco

Undergraduate Researcher

Dec 2025 – Present

Recife, BR

Supervisor: Cleber Zanchettin (UFPE)

- Developed a mechanistic interpretability framework for recursive neural architectures.
- Analyzed Jacobian spectra and attractor basins to characterize the dynamical behavior of learned representations.
- Investigated how training objectives influence algorithmic reasoning in recursive models compared with Transformer-based architectures.

National Institute of Science and Technology in Software Engineering (INES)

Undergraduate Research Assistant

Aug 2023 – May 2024

Remote

Supervisors: Kiev Gama (UFPE), Ana Paula Chaves (NAU).

- Developed quantitative analysis tools to evaluate accessibility in web and mobile applications for autistic adults.
- Contributed to accessibility research by analyzing usability and software quality metrics across mobile platforms.
- Co-authored peer-reviewed publications at IHC 2024 and SBSI 2025.

Extracurricular

President & Co-founder Ligia — Artificial Intelligence Academic League, Federal University of Pernambuco	Mar 2024 – Present Recife, Brazil
- Co-founded and currently lead an academic AI organization with approximately 40 undergraduate researchers. - Coordinate research and engineering projects spanning natural language processing, computer vision, and machine learning. - Organize technical seminars, study groups, and workshops. - Develop partnerships with universities, startups, and public institutions.	
Co-organizer II Advanced Summer School on Responsible AutoML	Nov 2025 Recife, Brazil
- Co-organized an international summer school jointly hosted by the Center for Informatics (UFPE) and the Faculty of Engineering of the University of Porto. - Coordinated partnerships and outreach with academic and industry organizations.	

Projects

F1Predict: Formula 1 Race Prediction System — [h](#)

Built an end-to-end Formula 1 race prediction system, from data collection and feature engineering to model deployment. Developed a FastAPI and PostgreSQL backend for model serving, achieving a 1-second prediction error.
Technologies: Python, PyTorch, scikit-learn, FastAPI, PostgreSQL.

MARS: Multi Agent Recommendation System — [vitornegromonte/mars](#)

Built a FastAPI-based multi-agent system that automates ArXiv paper collection, LLM-driven review, and recommendation.
Technologies: Python, FastAPI, LangChain.

Parkinson Diagnosis using Computer Vision: Campus Party'24 Keynote — [vitornegromonte/CPNE2024](#)

Developed a CNN for Parkinson's disease detection from spiral-drawing images, achieving 95% accuracy and outperforming state-of-the-art methods by approximately 10%.
Technologies: Python, PyTorch, OpenCV, Optuna, Zeus.

Publications

- [2] "A Mapping Review to Understand Web and Mobile Apps Accessibility for Adults with Autism", SBSI – 2025
- [1] "A Comparative Study on Accessibility for Autistic Individuals with Urban Mobility Apps", IHC – 2024

Awards & Grants

Cohere Labs Catalyst Grants Program	2025
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Teaching

Federal University of Pernambuco	
IF867 – Introduction to Deep Learning	2024 – 2026
IF866 – Computational Creativity	2024 – 2025

Skills

Programming – Python, C++, C, SQL, R, Bash
Machine Learning – PyTorch, TensorFlow, MLFlow, Weights & Biases, Keras, PyTorch Lightning
Backend & Data – FastAPI, PostgreSQL, Redis
Infrastructure – Docker, Linux, Git, MinIO
Languages – Portuguese (Native), English (C1)